Legendre–Rvachev Self-Reconstruction on $[-1, 1]$

Finite literal-shift synthesis, orthogonal atom blocks, positive rational energy series, infinite sinc integrals, and quarter-grid lifting identities

Research report for Vladimir Reshetnikov's `ProveIt` project

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Abstract

Let $u = \mathrm{up}$ be Rvachev's even C^∞ probability density supported on $[-1, 1]$. The current `ProveIt` corpus contains two apparently opposite constructions: every polynomial on a compact interval can be synthesized exactly from finitely many translates of u, while u itself has an absolutely and uniformly convergent Fourier–Legendre expansion $u = \sum_{n \geq 0} u_n P_{2n}$. Specializing the polynomial theorem gives, with $Q_d = \mathsf{M}(D)^{-1} P_d$,

$$P_d(x) = 2^{-d} \sum_{|k|<2^{d+1}} Q_d(k/2^d) u(x - k/2^d), \qquad -1 \leq x \leq 1.$$

For the even modes in the expansion this becomes a nested quarter-grid formula with scale $m_n = 4^n$. Substitution closes the loop blockwise:

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k|<2\cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n).$$

The basic loop and its non-flattening obstruction are already present in a recent repository report. The main contribution here is the additional structure exposed after closing it.

Each finite translate train is exactly the orthogonal mode $u_n P_{2n}$. Consequently the self-reconstruction is an orthogonal decomposition into finite cancellation blocks, with exact Pythagorean tails and a hierarchy of Sturm–Liouville energies. In particular, if $A_2 = \int_0^1 F(x)^2 \, \mathrm{d}x$, then

$$A_2 = \sum_{n=0}^{\infty} \frac{u_n^2}{4n+1} = \frac{1}{2\pi} \int_0^\infty \prod_{r=1}^{\infty} \left(\frac{\sin(t/2^r)}{t/2^r} \right)^2 \mathrm{d}t,$$

a positive series with exact rational partial sums; numerically $A_2 \approx 0.40443676798398526$.

The lattices $4^n \mathbb{Z}^{-1}$ are nested. Comparing two admissible lattices gives an explicit finite null train

$$\sum_{|k|<2rm} \left(1 - r\mathbf{1}_{r|k}\right) Q(k/(rm)) u(x - k/(rm)) = 0,$$

and, for successive Legendre partial sums, an exact lifting law: the coarse-to-fine coefficient update is the sum of a null-space gauge correction and one new orthogonal Legendre detail. Endpoint differentiation yields finite identities for all endpoint jets, while evaluation at the center gives an exact central-binomial dyadic sum rule.

Finally, combining the known factorial asymptotic of $Q_{2n}(0)$ with the nonanalyticity of u sharpens the existing non-flattening theorem. If $c_n^{(0)} = u_n Q_{2n}(0)/4^n$ is the repeated central-atom coefficient, then

$$\limsup_{n\to\infty} \frac{|c_n^{(0)}|^{1/n}}{n^2} = \frac{1}{\pi^2 \mathrm{e}^2}.$$

Thus the canonical atomwise coefficients are superexponential along a subsequence even though the resulting orthogonal blocks tend rapidly to zero. The archive includes exact rational verification, numerical coefficient scans, all figures, and a fully commented standalone Python program.

Novelty and evidence boundary

The Fourier product, probability model, rational moments, polynomial synthesis theorem, Fourier–Legendre coefficients, basic finite Legendre atomization, blockwise self-loop, fixed-scale partial sums, reciprocal-MGF pole asymptotics, and the original non-flattening theorem are imported from the repository snapshot `faa3a9b94ac0e71abdc53c36fdf428222e4d2a8c`. The recursive TeX tree was audited, with canonical consolidated sources close-read and archived duplicates or superseded siblings cross-checked by path inventory and targeted full-text searches. Results labeled "new here" were not found in that snapshot after searches for their defining formulas and terminology. This is a corpus-relative novelty statement, not an assertion of worldwide priority. Conjectures are explicitly labeled; numerical evidence is never used in place of a proof.

Formal status: Legendre energy

Editorial update (2026-08-29): at compiled source checkpoint `ca66a1247`, the subsequent Lean module `FabiusLegendreEnergy.lean` defines the polynomial-form blocks $B_n = u_n P_{2n}$ and formalizes their complete orthogonality formula (5.1), the coefficient Parseval identity (5.3), the exact coefficient tail (5.2), and the real-variable Legendre equality (6.3). Its public tail API includes both a `HasSum` theorem and a `tsum` equality.

Formal status: rational and Fourier energy

At compiled source checkpoint `9d5f41c2c`, the further module `FabiusLegendreRationalEnergy.lean` adds the three executable definitions `canonicalRvachevLegendreCoefficientRat`, `fabiusSquareEnergyTermRat`, and `fabiusSquareEnergyPartialSumRat`. Its fifteen public theorems are `canonicalRvachevLegendreCoefficientRat_cast`, `canonicalRvachevLegendreCoefficientRat_zero`, `fabiusSquareEnergyTermRat_cast`, `fabiusSquareEnergyTermRat_nonneg`, `fabiusSquareEnergyTermRat_zero`, `fabiusSquareEnergyPartialSumRat_cast`, `monotone_fabiusSquareEnergyPartialSumRat`, `fabiusSquareEnergyPartialSumRat_pos`, `fabiusSquareEnergyPartialSumRat_zero`, `fabiusSquareEnergyPartialSumRat_one`, `fabiusSquareEnergyPartialSumRat_two`, `fabiusSquareEnergyPartialSumRat_three`, `hasSum_fabiusSquareEnergy_ratCast`, `fabiusSquareEnergy_eq_tsum_ratCast`, and `tendsto_fabiusSquareEnergyPartialSumRat_cast`. They certify the report's claim that every partial sum in (6.3) is a positive rational number and that the sequence is monotone with real casts converging to A_2. They also certify all four displayed values in (6.6): $1/4$, $7/18$, $3271/8100$, and $3246043/8037225$ at $N = 0, 1, 2, 3$, respectively.

At compiled source checkpoint `b9b240bc0`, the further module `FabiusSquareEnergyFourier.lean` exports no definitions and exactly four theorems: `integral_norm_sq_rvachevFourier_eq_two_mul_fabiusSquareEnergy`, `fabiusSquareEnergy_eq_integral_Ioi_norm_sq_rvachevFourier`, `fabiusSquareEnergy_eq_integral_Ioi_tprod_sinc_sq`, and `fabiusSquareEnergy_eq_scaled_integral_Ioi_tprod_sinc_sq`. The first two identify the full real-axis squared Fourier mass with $2A_2$ and the positive-half-line mass with A_2. The last two close exactly the Fourier-product and infinite-sinc equalities (6.4) and (6.5), including the $1/(2\pi)$ normalization and the shift from the zero-based product to the one-based product.

Formal status: moment–Appell foundation

At compiled source checkpoint `1f9ce6fd9`, the subsequent module `RvachevMomentAppell.lean` exports five public definitions: `rvachevRawMomentRat`, `rvachevReciprocalMomentRat`, `rvachevAppellPolynomialRat`, `rvachevAppellPolynomial`, and `rvachevDeconvolvedPolynomial`. Its sixteen public theorems are `rvachevRawMomentRat_zero`, `rvachevRawMomentRat_even`, `rvachevRawMomentRat_odd`, `rvachevReciprocalMomentRat_zero`, `binomialConv_rvachevRawMomentRat_reciprocal`, `rvachevReciprocalMomentRat_eq_completeBellPolynomial`, `monic_rvachevAppellPolynomialRat`, `natDegree_rvachevAppellPolynomialRat`, `rvachevAppellPolynomial_eq_poly_cast`, `monic_rvachevAppellPolynomial`, `natDegree_rvachevAppellPolynomial`, `eval_rvachevAppellPolynomial_add`, `integral_pow_mul_rvachev_eq_rvachevRawMomentRat_cast`, `integral_eval_rvachevAppellPolynomial_add_mul_rvachev`, `natDegree_rvachevDeconvolvedPolynomial_le`, and `integral_eval_rvachevDeconvolvedPolynomial_add_mul_rvachev`. They formalize the full rational raw-moment sequence, its formal binomial-convolution reciprocal and complete-Bell description, rational and real monic Appell families of exact degree, and the polynomial smoothing inverse used below. This closes the formal coefficient content behind (2.10) and the operational polynomial-deconvolution content of (2.12)–(2.13), but not the displayed analytic reciprocal-MGF, differential-series, or Appell generating-function identities.

Formal status: synthesis and translate blocks

At compiled source checkpoint `c51a41fcf`, `RvachevPolynomialSynthesis.lean` exports no public definitions and exactly four public theorems: `tsum_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`, `normalized_tsum_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`, `sum_Ioo_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`, and `normalized_sum_Ioo_rvachevDeconvolvedPolynomial_mul_shifted_rvachevUp`. They prove global raw and normalized `tsum` synthesis for every nonzero natural mesh m with $\deg p \le \nu_2(m)$, and on $[-1, 1]$ its exact finite open-index form $-2m < k < 2m$. Thus the sufficient-condition and boxed-formula content of (2.14) is formalized.

At the same checkpoint, `FabiusLegendreTranslateBlocks.lean` exports six public definitions: `rvachevLegendreDeconvolutionPolynomial`, `rvachevLegendreScale`, `rvachevLegendreIndexSet`, `rvachevLegendreAtomCoefficient`, `rvachevLegendreTranslateBlock`, and `rvachevTranslateGram`. Its five public theorems are `eval_legendrePolynomial_eq_sum_rvachevUp`, `eval_legendrePolynomial_even_eq_sum_rvachevUp`, `rvachevLegendreTranslateBlock_eq_rvachevLegendreBlock`, `intervalIntegral_rvachevLegendreTranslateBlock_mul`, and `sum_rvachevLegendreAtomCoefficient_mul_gram`. They formalize the deconvolved Legendre definition (3.1), the finite formulas (3.12) and (3.17), the literal finite block data and equality (4.5)–(4.6), complete orthogonality of those translate blocks, and the atom-Gram identity (5.5).

The new APIs do not prove minimality or sharpness of the displayed mesh; reciprocal/deconvolution parity or the derivative, Gegenbauer, Appell, and low-degree formulas for Q_d; rationality of the atom rows; a named outer translate-block `HasSum` for (4.7); the one-mesh partial-sum identity (4.9); or the later refinement, projector, and asymptotic layers.

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1 Corpus audit and mathematical orientation

1.1 Principal repository inputs

The relevant corpus is unusually rich and contains several reports answering closely related prompts. The following sources carry the load-bearing facts used below.

Repository source	**Role in this report**
`Fabius_Function_and_Rvachev_Up/`	Canonical normalization, Fourier product, moments, dyadic values, and the exact Fourier–Legendre expansion with rational coefficients.
`Up_Polynomial_Synthesis/`	Consolidated exact reproduction of arbitrary polynomials by shifted or dilated up-atoms; dyadic alias multiplicities; Bell–Bernoulli deconvolution; local nullspaces and endpoint compression.
`lagrange_rvachev_loop_report_v3/`	The directly preceding Legendre-aware loop report: deconvolved Legendre polynomials, Christoffel–Darboux synthesis, fixed-scale partial loops, factorial growth of $Q_{2n}(0)$, and rigorous non-flattening.
The other four Lagrange–Rvachev loop siblings	Independent checks of the same general finite synthesis, including exact right-inverse calculations and oblique-projector conditioning. They were used as consistency checks rather than treated as five independent theorems.
Fourier-decay, inverse, Lambert-W, q-analog, and asymptotic reports	Context for the sinc product, dyadic pole shells, lower/principal Lambert branches, q-binomial formulas, and proposed asymptotics of endpoint and spectral quantities.

The recursive tree also contains historical versions, source copies inside archives, and generated consolidation documents. Counting every byte-distinct copy as independent evidence would obscure rather than strengthen the audit. The approach taken here was therefore: enumerate the full TeX tree, identify canonical or newest representatives, compare neighboring reports, and search the entire repository for the precise formulas introduced below.

1.2 Imported, specialized, and new deductions

Status	Mathematical content
Imported	$\widehat{u}(\xi) = \prod_{j\geq 0} \operatorname{sinc}_\pi(\xi/2^j)$, $\operatorname{ord}_{\xi=m} \widehat{u} = 1+\nu_2(m)$, the MGF and cumulants, rational moments, and exact dyadic values.
Imported	Exact polynomial synthesis at any admissible dyadic oversampling ratio, including the unit-radius literal-shift formula on $[-1,1]$.
Imported	$u = \sum u_n P_{2n}$ with absolute and uniform convergence; exact rational u_n; $Q_n = \mathscr{D}_u P_n$; the basic block loop and fixed-scale partial loops.
Imported	The factorial asymptotic of $Q_{2n}(0)$ and the theorem that the canonical loop cannot be flattened into an absolute or unconditional atom series.
Specialized here	A self-contained Legendre specialization for every P_d, including explicit Bell–Bernoulli coefficient formulas, low-degree examples, and endpoint/center sum rules.
New here, corpus-relative	Orthogonality of the finite translate blocks, atom-Gram identities, exact Pythagorean tails, rank-one atomized detail projectors, and the full Sturm–Liouville energy hierarchy.
New here, corpus-relative	The positive rational expansion and infinite-sinc integral for $\int_0^1 F^2$, together with a nested exact dyadic-Fabius square series.
New here, corpus-relative	General interlevel null trains, the quarter-grid refinement cocycle, and an exact lifting decomposition of successive fixed-scale coefficient vectors.
New here, corpus-relative	Even-projector dual-frame trace identities and the sharp root-limsup law $\limsup |c_n^{(0)}|^{1/n}/n^2 = (\pi^2 e^2)^{-1}$.

Table 2: Status of the principal results.

2 Rvachev's atom, its MGF, and polynomial deconvolution

2.1 Normalization and Fourier product

Let $u = \mathrm{up}$ be the unique even nonnegative C^∞ function such that

$$\operatorname{supp} u = [-1,1], \qquad u(0) = 1, \qquad u'(x) = 2u(2x+1) - 2u(2x-1). \tag{2.1}$$

The chosen normalization also gives

$$\int_{-1}^{1} u(x)\,\mathrm{d}x = 1. \tag{2.2}$$

With $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\,\mathrm{d}x$,

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_\pi(\xi/2^j), \qquad \operatorname{sinc}_\pi(t) = \frac{\sin(\pi t)}{\pi t}. \tag{2.3}$$

Every nonzero integer zero has the exact multiplicity

$$\operatorname{ord}_{\xi=m} \widehat{u}(\xi) = 1 + \nu_2(m). \tag{2.4}$$

These multiplicities are the source of the sharp dyadic reproduction order. They are also the Fourier shadow of the q-integer and Thue–Morse annihilators appearing in local coefficient compression.

Equivalently, if V_j are independent uniform random variables on $[-1,1]$, then

$$Z = \sum_{j=1}^{\infty} 2^{-j} V_j \tag{2.5}$$

has density u. Hence the moment generating function is

$$\mathsf{M}(z) = \mathbb{E} e^{zZ} = \int_{-1}^{1} e^{zx} u(x)\,\mathrm{d}x = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \tag{2.6}$$

2.2 Bernoulli cumulants and Bell-polynomial inverse

Write

$$\log \mathsf{M}(z) = \sum_{r\ge 1} \kappa_r \frac{z^r}{r!}. \tag{2.7}$$

Symmetry gives $\kappa_{2r+1} = 0$, while

$$\kappa_{2r} = \frac{B_{2r}}{2r(1-4^{-r})} = \frac{2^{2r-1}B_{2r}}{r(2^{2r}-1)}. \tag{2.8}$$

Define reciprocal coefficients by

$$\frac{1}{\mathsf{M}(z)} = \sum_{r=0}^{\infty} \beta_r \frac{z^r}{r!}. \tag{2.9}$$

Then $\beta_{2r+1} = 0$ and, in complete Bell-polynomial notation,

$$\beta_n = Y_n(-\kappa_1, -\kappa_2, \dots, -\kappa_n). \tag{2.10}$$

The first nonzero values are

$$\beta_0 = 1, \qquad \beta_2 = -\frac{1}{9}, \qquad \beta_4 = \frac{31}{675}, \qquad \beta_6 = -\frac{2347}{59535}, \qquad \beta_8 = \frac{1904369}{32531625}. \tag{2.11}$$

On polynomials the formal operator

$$\mathscr{D}_u := \mathsf{M}(D)^{-1} = \sum_{r\ge 0} \beta_r \frac{D^r}{r!} \tag{2.12}$$

terminates exactly. It reverses convolution by the up probability law in the polynomial category. Its Appell sequence

$$A_n(x) = \mathscr{D}_u x^n, \qquad \frac{e^{xz}}{\mathsf{M}(z)} = \sum_{n\ge 0} A_n(x) \frac{z^n}{n!} \tag{2.13}$$

is therefore a finite Bell–Bernoulli coefficient engine for every synthesis formula below.

2.3 The imported exact polynomial theorem

The unit-radius specialization of the repository's polynomial synthesis result is the following.

Theorem 2.1 (Exact literal-shift synthesis; imported)**.** *Let p be a polynomial of degree at most d. If $m \in \mathbb{N}$ satisfies $\nu_2(m) \ge d$, put $p^\sharp = \mathscr{D}_u p$. Then, for $-1 \le x \le 1$,*

$$\boxed{p(x) = \frac{1}{m} \sum_{|k|<2m} p^\sharp(k/m) u(x - k/m).} \tag{2.14}$$

The canonical minimal admissible ratio is $m = 2^d$, giving $4m-1$ displayed atoms.

Mechanism of proof. Poisson summation converts the sampled translate train into a sum over Fourier aliases. At the zero alias, convolution by u acts on polynomials as $\mathsf{M}(D)$, which is exactly inverted by $\mathscr{D}_u$. At every nonzero alias, the zero multiplicity (2.4) is at least $d+1$, so all distributional derivatives through degree d vanish. Compact support reduces the globally locally finite train to the displayed finite range on $[-1, 1]$. The leading coefficient of a degree-d polynomial shows why the condition $\nu_2(m) \ge d$ is sharp within this stationary Poisson-synthesis scheme. ☐

3 Finite shifted-up synthesis of Legendre polynomials

3.1 The deconvolved Legendre family

Let P_d be the Legendre polynomial normalized by $P_d(1) = 1$, and define

$$Q_d(x) := \mathscr{D}_u P_d(x). \tag{3.1}$$

Because $\mathscr{D}_u$ contains only even derivatives,

$$Q_d(-x) = (-1)^d Q_d(x). \tag{3.2}$$

There are several useful finite forms. Directly from (2.12),

$$Q_d(x) = \sum_{r=0}^{\lfloor d/2 \rfloor} \frac{\beta_{2r}}{(2r)!} P_d^{(2r)}(x). \tag{3.3}$$

Using $P_d^{(2r)} = (4r-1)!!\, C_{d-2r}^{2r+1/2}$ gives

$$Q_d(x) = \sum_{r=0}^{\lfloor d/2 \rfloor} \frac{\beta_{2r}(4r-1)!!}{(2r)!} C_{d-2r}^{2r+1/2}(x). \tag{3.4}$$

Alternatively, applying $\mathscr{D}_u x^j = A_j(x)$ to the standard power expansion,

$$Q_d(x) = 2^{-d} \sum_{r=0}^{\lfloor d/2 \rfloor} (-1)^r \frac{(2d-2r)!}{r!(d-r)!(d-2r)!} A_{d-2r}(x). \tag{3.5}$$

Thus every coefficient is an explicit finite Bell polynomial in the Bernoulli numbers.

The first members are

$$Q_0(x) = 1, \qquad Q_1(x) = x, \tag{3.6}$$
$$Q_2(x) = \frac{3}{2}x^2 - \frac{2}{3}, \tag{3.7}$$
$$Q_3(x) = \frac{5}{2}x^3 - \frac{7}{3}x, \tag{3.8}$$
$$Q_4(x) = \frac{35}{8}x^4 - \frac{20}{3}x^2 + \frac{134}{135}, \tag{3.9}$$
$$Q_5(x) = \frac{63}{8}x^5 - \frac{35}{2}x^3 + \frac{33}{5}x, \tag{3.10}$$
$$Q_6(x) = \frac{231}{16}x^6 - \frac{175}{4}x^4 + \frac{889}{30}x^2 - \frac{1426}{567}. \tag{3.11}$$

They are not orthogonal. Their role is instead exact coefficient preconditioning: Q_d removes the moments introduced by the atom u.

3.2 The requested finite Legendre representation

Theorem 3.1 (Finite shifted-up representation of P_d)**.** *For every $d \geq 0$, put $m_d = 2^d$. Then*

$$\boxed{P_d(x) = \frac{1}{2^d} \sum_{|k|<2^{d+1}} Q_d(k/2^d) u(x - k/2^d), \qquad -1 \leq x \leq 1.} \tag{3.12}$$

The formula has $2^{d+2} - 1$ literal translates of the same, unscaled up-function. Within the stationary scheme of theorem 2.1, $m_d = 2^d$ is minimal.

Proof. Apply theorem 2.1 to $p = P_d$. Its exact degree is d and its deconvolution is Q_d, so the canonical ratio is $m = 2^d$. □

The first three cases are already informative. On $[-1, 1]$,

$$1 = u(x+1) + u(x) + u(x-1), \tag{3.13}$$

$$x = \frac{1}{4}\sum_{k=-3}^{3} k\, u(x - k/2), \tag{3.14}$$

$$P_2(x) = \sum_{k=-7}^{7} \frac{9k^2 - 64}{384}\, u(x - k/4). \tag{3.15}$$

The signs in (3.15) are indispensable: exact polynomial reproduction is a cancellation identity, not a positive-mixture formula.

3.3 Even modes and the nested quarter-grid

For the modes that occur in the expansion of the even function u, set

$$m_n := 2^{2n} = 4^n. \tag{3.16}$$

Then (3.12) becomes

$$\boxed{P_{2n}(x) = \frac{1}{4^n}\sum_{|k|<2\cdot 4^n} Q_{2n}(k/4^n)u(x - k/4^n).} \tag{3.17}$$

The grids are nested:

$$\frac{1}{m_n}\mathbb{Z} \subset \frac{1}{m_{n+1}}\mathbb{Z}, \qquad m_{n+1} = 4m_n. \tag{3.18}$$

This elementary observation is the seed of the exact multiresolution identities in section 8.

4 Closing the existing Fourier–Legendre loop

4.1 Exact rational Legendre coefficients

The canonical expansion in the repository is

$$u(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x), \qquad u_n = \frac{4n+1}{2}\int_{-1}^{1} u(x)P_{2n}(x)\,\mathrm{d}x. \tag{4.1}$$

It converges absolutely and uniformly on $[-1, 1]$. If $\mu_{2j} = \int x^{2j}u(x)\,\mathrm{d}x$, then

$$u_n = \frac{4n+1}{2^{2n+1}}\sum_{j=0}^{n}(-1)^{n+j}\binom{2n}{n+j}\binom{2n+2j}{2n}\mu_{2j}. \tag{4.2}$$

The repository's dyadic moment bridge

$$\mu_{2j} = 2(2j)!\,2^{\binom{2j+1}{2}}F(2^{-2j-1}) \tag{4.3}$$

makes every u_n rational. Combining the two equations gives

$$u_n = 4^{-n}(4n+1)\sum_{j=0}^{n}(-1)^{n+j}\binom{2n}{n+j}\binom{2n+2j}{2n}(2j)!2^{\binom{2j+1}{2}}F(2^{-2j-1}). \tag{4.4}$$

n	u_n	$Q_{2n}(0)$	$u_nQ_{2n}(0)/4^n$
0	$1/2$	1	$1/2$
1	$-5/6$	$-2/3$	$5/36$
2	$11/30$	$134/135$	$737/32400$
3	$143/5670$	$-1426/567$	$-101959/102876480$
4	$-7291/85050$	$7587914/722925$	$-0.00351481095499\ldots$
5	$99137/2485890$	$-9200178166/130509225$	$-0.00274542090064\ldots$
6	$-5726162257/213774110550$	$728.294733513\ldots$	$-0.00476272782254\ldots$

Table 3: First Legendre, deconvolution, and central-atom coefficients. Exact fractions for all displayed decimals are in the accompanying CSV file.

4.2 The blockwise self-loop

Substituting (3.17) into (4.1) yields the exact loop already established in the recent Legendre-aware repository report.

Theorem 4.1 (Blockwise Legendre–Rvachev loop; imported)**.** *Define*

$$B_n(x) := \frac{u_n}{m_n} \sum_{|k|<2m_n} Q_{2n}(k/m_n)u(x - k/m_n). \tag{4.5}$$

Then, on $[-1, 1]$,

$$B_n(x) = u_nP_{2n}(x), \tag{4.6}$$

and

$$\boxed{u(x) = \sum_{n=0}^{\infty} B_n(x).} \tag{4.7}$$

The outer series converges absolutely and uniformly when every finite inner train is retained as one block.

Proof. Equation (4.6) is (3.17) multiplied by u_n. Since $\|P_j\|_{L^\infty[-1,1]} \le 1$ and the imported expansion satisfies $\sum |u_n| < \infty$, the Weierstrass test proves the stated block convergence. □

The grouping qualification is essential. The same translate $u(x)$ occurs as the $k = 0$ atom in every block, and its scalar coefficients eventually become large. The existing non-flattening theorem proves that the double series is neither absolutely nor unconditionally convergent when the individual atoms are detached from their blocks. We sharpen that conclusion quantitatively in section 11.

4.3 One lattice for each partial sum

Let

$$S_N(x) = \sum_{n=0}^{N} u_nP_{2n}(x), \qquad C_N(t) = \mathscr{D}_uS_N(t) = \sum_{n=0}^{N} u_nQ_{2n}(t). \tag{4.8}$$

Since $\deg S_N = 2N$, the whole partial sum can be synthesized on the single finest grid $m_N = 4^N$:

$$\boxed{S_N(x) = \frac{1}{m_N} \sum_{|k|<2m_N} C_N(k/m_N)u(x - k/m_N).} \tag{4.9}$$

All coefficients are rational. This representation is the input for the coarse-to-fine lifting law below.

5 Orthogonal finite translate blocks

The closed loop becomes substantially more rigid when viewed in $L^2[-1,1]$. Although the atoms within a block are highly nonorthogonal, the *finite trains themselves* are exactly orthogonal.

Theorem 5.1 (Orthogonal self-translate blocks; new here)**.** *For* $n, m \geq 0$*,*

$$\boxed{\langle B_n, B_m\rangle_{L^2[-1,1]} = \delta_{nm}\frac{2u_n^2}{4n+1}.} \tag{5.1}$$

Consequently,

$$\|u - S_N\|_2^2 = \sum_{n>N} \frac{2u_n^2}{4n+1}, \tag{5.2}$$

and

$$\|u\|_2^2 = \sum_{n=0}^{\infty} \frac{2u_n^2}{4n+1}. \tag{5.3}$$

Proof. By (4.6), $B_n = u_n P_{2n}$ on the interval. Legendre orthogonality gives

$$\int_{-1}^{1} P_{2n}(x)P_{2m}(x)\,\mathrm{d}x = \delta_{nm}\frac{2}{4n+1}.$$

This proves (5.1). The exact tail and Parseval identity follow from orthogonal convergence of (4.7). □

5.1 An exact atom-Gram identity

Write

$$a_{n,k} := \frac{u_n}{m_n} Q_{2n}(k/m_n), \qquad g_{n,k}(x) := u(x - k/m_n)\big|_{[-1,1]}. \tag{5.4}$$

Then $B_n = \sum_k a_{n,k} g_{n,k}$. Expanding (5.1) gives a finite, exact identity for the cross-scale Gram matrices:

$$\boxed{\sum_{|k|<2m_n} \sum_{|\ell|<2m_m} a_{n,k} a_{m,\ell} \int_{-1}^{1} u(x - k/m_n)u(x - \ell/m_m)\,\mathrm{d}x = \delta_{nm}\frac{2u_n^2}{4n+1}.} \tag{5.5}$$

Thus complicated cross-scale translate correlations cancel exactly whenever their coefficient rows arise from distinct Legendre modes.

Remark 5.2. Equation (5.5) does not say that the individual translates form an orthogonal wavelet system. They do not. It says that a very particular signed finite combination on each scale diagonalizes the Legendre inner product. The orthogonality lives at block level.

5.2 Rank-one atomized detail projectors

For $f \in L^2[-1,1]$, define

$$\Delta_n f = \frac{4n+1}{2} \langle f, P_{2n}\rangle P_{2n}. \tag{5.6}$$

Theorem 5.3 (Exact atomized spectral details; new here)**.** *Each* Δ_n *is the orthogonal rank-one projector onto* $\operatorname{span}\{P_{2n}\}$ *and admits the finite atom realization*

$$\boxed{(\Delta_n f)(x) = \frac{4n+1}{2m_n} \langle f, P_{2n}\rangle \sum_{|k|<2m_n} Q_{2n}(k/m_n)u(x - k/m_n).} \tag{5.7}$$

Moreover,

$$\Delta_n^* = \Delta_n, \qquad \Delta_n\Delta_m = \delta_{nm}\Delta_n, \qquad \Delta_n u = B_n. \tag{5.8}$$

Proof. The first two identities are standard Legendre orthogonality. Substitute (3.17) into (5.6) to obtain (5.7). Finally, $u_n = (4n+1)\langle u, P_{2n}\rangle/2$. ☐

Hence the even Legendre projector

$$\Pi_N^{\mathrm{e}} := \sum_{n=0}^{N} \Delta_n \tag{5.9}$$

has an exact decomposition into $N+1$ mutually orthogonal finite translate blocks. This is stronger than merely saying that the partial sum is a best approximation: it identifies an exact finite-atom realization of every spectral detail operator.

6 A positive rational energy series and an infinite sinc integral

6.1 The square energy of the Fabius function

On $[-1,1]$ the usual Fabius function and the even up-function are related by

$$u(x) = F(1-|x|). \tag{6.1}$$

Define

$$A_2 := \int_0^1 F(t)^2\,\mathrm{d}t. \tag{6.2}$$

Then $\|u\|_2^2 = 2A_2$.

Theorem 6.1 (Positive rational Legendre series for the energy; new here). *The Fabius square energy has the exact representations*

$$A_2 = \boxed{\sum_{n=0}^{\infty} \frac{u_n^2}{4n+1}} \tag{6.3}$$

$$= \int_0^{\infty} \prod_{j=0}^{\infty} \operatorname{sinc}_\pi^2(\xi/2^j)\,\mathrm{d}\xi \tag{6.4}$$

$$= \frac{1}{2\pi}\int_0^{\infty} \prod_{r=1}^{\infty} \left(\frac{\sin(t/2^r)}{t/2^r}\right)^2 \mathrm{d}t. \tag{6.5}$$

Every partial sum in (6.3) is a positive rational number and increases monotonically to A_2.

Proof. Divide (5.3) by two and use (6.1); this gives (6.3). Each u_n is rational by (4.4). Parseval's theorem for the Fourier convention (2.3) gives

$$\|u\|_2^2 = \int_{\mathbb{R}} |\widehat{u}(\xi)|^2\,\mathrm{d}\xi.$$

Evenness and division by two yield (6.4). Finally substitute $t = 2\pi\xi$. ☐

The first exact partial sums are

$$A_{2,0} = \frac{1}{4}, \qquad A_{2,1} = \frac{7}{18}, \qquad A_{2,2} = \frac{3271}{8100}, \qquad A_{2,3} = \frac{3246043}{8037225}. \tag{6.6}$$

The accompanying exact calculation gives

$$A_2 \approx 0.40443676798398526. \tag{6.7}$$

The displayed value is a stabilized numerical evaluation of a long positive rational partial sum, not a claim that the last digit has been certified by an independent tail enclosure.

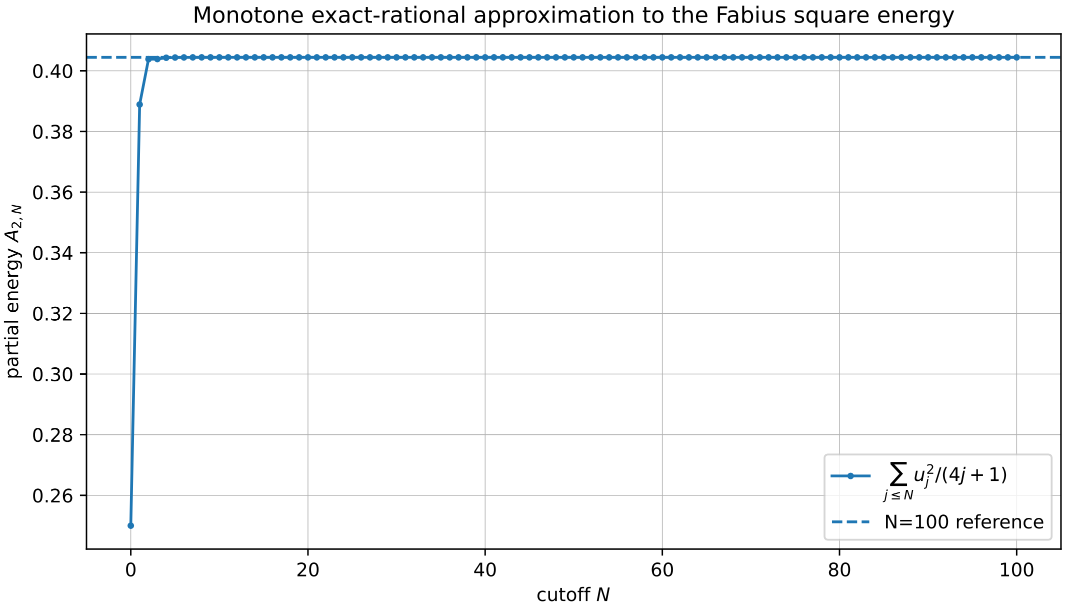

Figure 1: Monotone exact-rational partial sums of $A_2 = \sum u_n^2/(4n+1)$. The dashed line is the $N = 100$ computed reference.

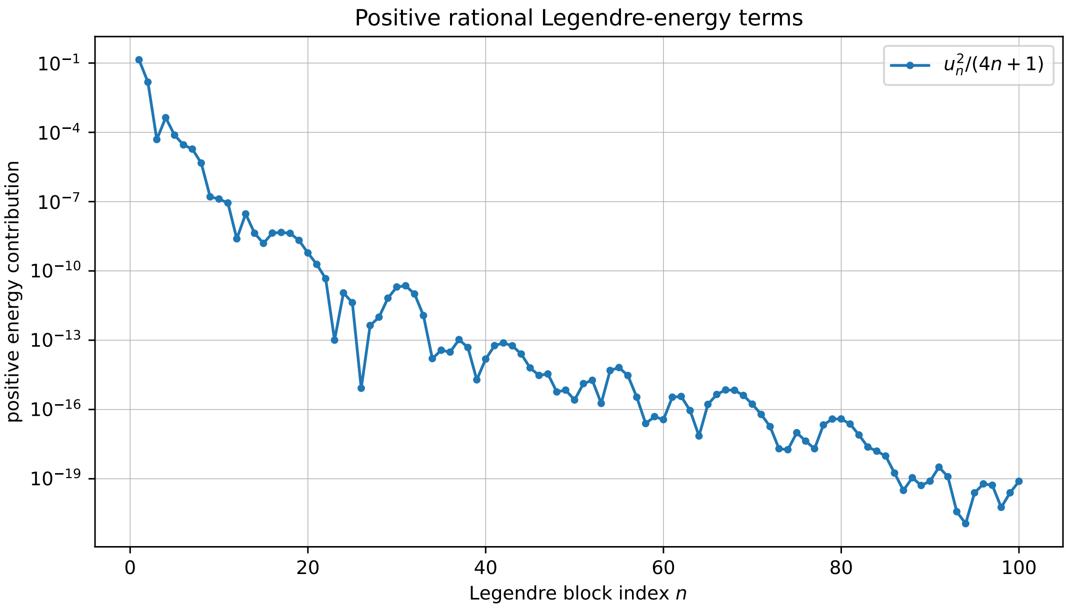

Figure 2: Positive contributions $u_n^2/(4n+1)$ on a logarithmic scale.

6.2 A nested dyadic-Fabius square series

Let

$$R_n := \sum_{j=0}^{n} (-1)^{n+j} \binom{2n}{n+j} \binom{2n+2j}{2n} (2j)!\,2^{\binom{2j+1}{2}} F(2^{-2j-1}). \tag{6.8}$$

Then $u_n = 4^{-n}(4n+1)R_n$, and theorem 6.1 immediately gives

$$\boxed{A_2 = \sum_{n=0}^{\infty}(4n+1)16^{-n}R_n^2.} \tag{6.9}$$

Every summand is the square of a finite rational combination of dyadic Fabius values. Substituting any of the repository's q-binomial or Thue–Morse formulas for those values turns (6.9) into a completely finite-inside-infinite q-arithmetic series.

Problem 6.2 (Arithmetic of the square energy)**.** *Determine whether A_2 is irrational or transcendental, and whether the three representations* (6.3)–(6.5) *admit effective Diophantine approximations stronger than their obvious rational partial sums. The positivity of* (6.3) *is potentially useful because it eliminates cancellation from the approximation error.*

7 Sturm–Liouville and Sobolev energy hierarchies

Let

$$\mathcal{L}f = -\frac{\mathrm{d}}{\mathrm{d}x}\left((1-x^2)f'(x)\right). \tag{7.1}$$

Then

$$\mathcal{L}P_{2n} = \lambda_n P_{2n}, \qquad \lambda_n = 2n(2n+1). \tag{7.2}$$

The endpoint factor $1-x^2$ makes integration by parts compatible with the natural Legendre boundary condition.

Theorem 7.1 (Orthogonal derivative energies; new here)**.** *For $n, m \geq 0$,*

$$\int_{-1}^{1}(1-x^2)B_n'(x)B_m'(x)\,\mathrm{d}x = \delta_{nm}\lambda_n\frac{2u_n^2}{4n+1}. \tag{7.3}$$

More generally, for every real $s \geq 0$ for which the spectral norm is defined,

$$\boxed{\left\|(I+\mathcal{L})^{s/2}u\right\|_2^2 = \sum_{n=0}^{\infty}(1+\lambda_n)^s\frac{2u_n^2}{4n+1}.} \tag{7.4}$$

In particular,

$$\int_{-1}^{1}(1-x^2)|u'(x)|^2\,\mathrm{d}x = \sum_{n=0}^{\infty}\lambda_n\frac{2u_n^2}{4n+1}. \tag{7.5}$$

Proof. Use $B_n = u_nP_{2n}$ and self-adjointness:

$$\int(1-x^2)B_n'B_m' = \langle\mathcal{L}B_n, B_m\rangle = \lambda_n\langle B_n, B_m\rangle.$$

Now apply theorem 5.1. Equation (7.4) is the spectral theorem for the nonnegative self-adjoint Legendre operator, and (7.5) is its $s=1$ quadratic-form part after subtracting (5.3). □

Because u is C^∞ and flat at the endpoints, all integer levels of this hierarchy are finite. The formulas therefore give infinitely many positive series built from the same rational coefficients u_n, weighted by explicit polynomial eigenvalues. Their partial sums are rational and monotone, providing convenient certified lower bounds for weighted derivative energies.

8 Nested grids, exact null trains, and a lifting law

8.1 A general interlevel null relation

For an admissible lattice m, define the finite synthesis operator

$$(T_m c)(x) := \sum_{|k|<2m} c_k u(x - k/m), \qquad -1 \le x \le 1. \tag{8.1}$$

If p is a polynomial and $Q = \mathscr{D}_u p$, its canonical coefficient vector is

$$(F_m p)_k := \frac{1}{m} Q(k/m). \tag{8.2}$$

Then theorem 2.1 says $T_m F_m p = p$.

Theorem 8.1 (Interlevel polynomial null train; new here)**.** *Let p have degree at most d, put $Q = \mathscr{D}_u p$, and suppose both m and rm are admissible, i.e. $\nu_2(m), \nu_2(rm) \ge d$. Then*

$$\boxed{\frac{1}{rm} \sum_{|k|<2rm} \big(1 - r\mathbf{1}_{r|k}\big) Q(k/(rm)) u(x - k/(rm)) = 0} \tag{8.3}$$

for every $x \in [-1, 1]$.

Proof. The fine representation is

$$p(x) = \frac{1}{rm} \sum_{|k|<2rm} Q(k/(rm)) u(x - k/(rm)).$$

Rewrite the coarse representation on the fine index set by setting $k = rj$:

$$p(x) = \frac{r}{rm} \sum_{\substack{|k|<2rm \\ r|k}} Q(k/(rm)) u(x - k/(rm)).$$

Subtract the two identities. □

For the even Legendre hierarchy, take $m = m_n = 4^n$ and $r = 4$:

$$\sum_{|k|<8m_n} \big(1 - 4\mathbf{1}_{4|k}\big) Q(k/(4m_n)) u(x - k/(4m_n)) = 0. \tag{8.4}$$

This is an explicit vector in the local kernel of the fine-grid synthesis operator. It is not obtained by solving a matrix nullspace: its entries are samples of the same deconvolved polynomial, multiplied by the fixed mask $1 - 4\mathbf{1}_{4|k}$.

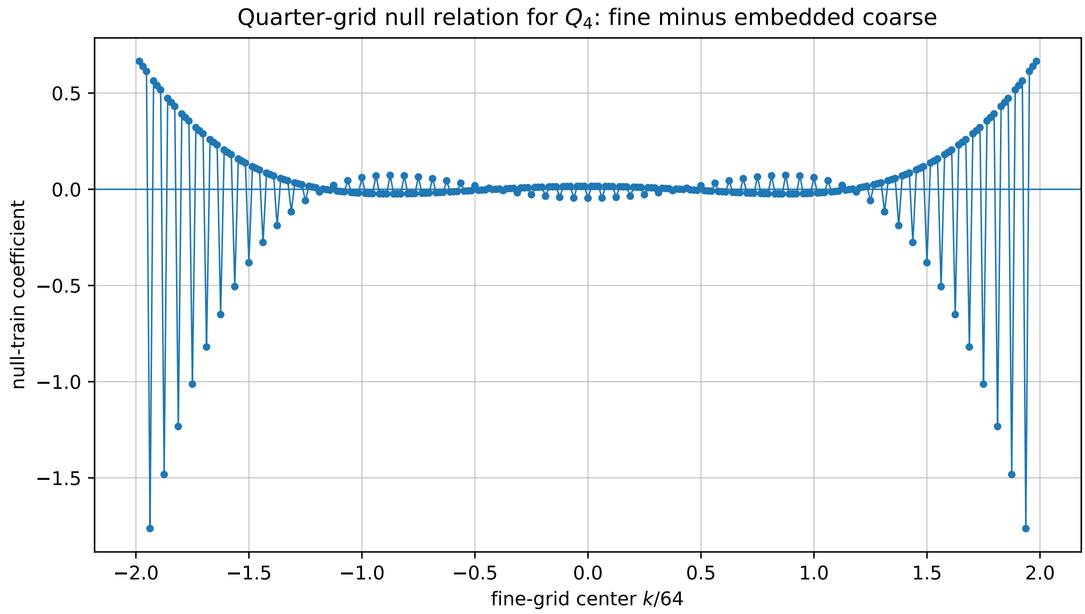

Figure 3: The exact quarter-grid null coefficients for Q_4, comparing $m = 16$ with $m = 64$. The corresponding translate train vanishes identically on $[-1, 1]$ despite its substantial coefficients.

8.2 The refinement cocycle

Let E_r embed a coarse vector into the r-times finer grid:

$$(E_r c)_k = \begin{cases} c_{k/r}, & r \mid k, \\ 0, & r \nmid k. \end{cases} \tag{8.5}$$

Then

$$G_{m;r}p := F_{rm}p - E_r F_m p \tag{8.6}$$

lies in $\ker T_{rm}$. Refinements compose by the exact cocycle identity

$$\boxed{G_{m;rs}p = G_{rm;s}p + E_s G_{m;r}p.} \tag{8.7}$$

Indeed, both sides telescope to $F_{rsm}p - E_s E_r F_m p$. Thus all multilevel null corrections can be built from one-step corrections without recomputing a kernel basis.

8.3 A quarter-grid lifting decomposition

Let $a^{(N)}$ be the canonical fixed-scale coefficient vector for S_N:

$$a_k^{(N)} = \frac{1}{m_N} C_N(k/m_N), \qquad |k| < 2m_N. \tag{8.8}$$

Embed it into level $N + 1$ with E_4. Since $C_{N+1} = C_N + u_{N+1} Q_{2N+2}$, direct subtraction yields the following exact law.

Theorem 8.2 (Exact Legendre–Rvachev lifting step; new here). *For $m = m_N$ and $|k| < 8m$,*

$$\boxed{a_k^{(N+1)} - (E_4 a^{(N)})_k = \frac{1 - 4\mathbf{1}_{4|k}}{4m} C_N(k/(4m)) + \frac{u_{N+1}}{4m} Q_{2N+2}(k/(4m)).} \tag{8.9}$$

The first term synthesizes the zero function on $[-1, 1]$; the second synthesizes exactly the new orthogonal detail $B_{N+1} = u_{N+1} P_{2N+2}$. Consequently,

$$T_{m_{N+1}}\left(a^{(N+1)} - E_4 a^{(N)}\right) = B_{N+1}. \tag{8.10}$$

Proof. At a fine-grid point $t = k/(4m)$,

$$a_k^{(N+1)} = \frac{1}{4m}\big(C_N(t) + u_{N+1}Q_{2N+2}(t)\big),$$

whereas $(E_4 a^{(N)})_k = 4\mathbf{1}_{4|k} C_N(t)/(4m)$. This proves (8.9). The first term is theorem 8.1 applied to $p = S_N$; the second is (3.17) multiplied by u_{N+1}. □

This is analogous in form to a lifting scheme, but with a significant qualification. The gauge component is not a wavelet detail: it is exactly invisible on the target interval. The genuine detail is the orthogonal Legendre mode. Coefficient refinement therefore decomposes into

new coefficients = embedded old coefficients + null-space gauge + orthogonal spectral detail.

The accompanying program verifies the complete $N = 1 \to 2$ identity, both coefficientwise and as a function, with exact rational residual zero.

9 Endpoint, center, and derivative sum rules

The finite synthesis identity can be sampled or differentiated to produce families of exact dyadic identities involving u and its derivatives.

Theorem 9.1 (Endpoint jet sum rules; new here). *Let $d \geq 0$, $m = 2^d$, and $0 \leq r \leq d$. Then*

$$\boxed{\frac{1}{m}\sum_{k=1}^{2m-1} Q_d(k/m)u^{(r)}(1-k/m) = \frac{1}{2^r}\frac{(d+r)!}{r!(d-r)!}}. \tag{9.1}$$

For $r > d$, the same left side is zero. At the left endpoint the value is $(-1)^{d+r}$ times the right-hand side.

Proof. Differentiate (3.12) r times. At $x = 1$, all terms with $k \leq 0$ vanish because their atom arguments are at or beyond the flat endpoint 1; the displayed range $1 \leq k \leq 2m-1$ is exactly the surviving support. The classical endpoint derivative is

$$P_d^{(r)}(1) = 2^{-r}\frac{(d+r)!}{r!(d-r)!}$$

for $r \leq d$, and zero afterward. Parity gives the left endpoint. □

The case $r = 0$ is particularly simple:

$$\sum_{k=1}^{2m-1} Q_d(k/m)u(1-k/m) = m. \tag{9.2}$$

All arguments are dyadic, so every term is rational; the same is true for all endpoint-jet identities because derivatives of u at dyadic points are rational [4].

Theorem 9.2 (Central-binomial dyadic sum rule; new here). *For $n \geq 0$ and $m = 4^n$,*

$$\boxed{Q_{2n}(0) + 2\sum_{k=1}^{m-1} Q_{2n}(k/m)u(k/m) = (-1)^n\binom{2n}{n}}. \tag{9.3}$$

Proof. Evaluate (3.17) at $x = 0$. The $k = \pm m$ boundary terms vanish, and parity of Q_{2n} and u pairs the remaining positive and negative indices. Finally,

$$mP_{2n}(0) = 4^n(-1)^n\frac{1}{4^n}\binom{2n}{n}.$$

□

This identity is notable because the factorially large value $Q_{2n}(0)$ is balanced by a finite rational dyadic sum to leave the merely exponential central binomial coefficient. It is a local, exact manifestation of the cancellation that dominates the closed loop.

9.1 Mode-wise biorthogonality

Multiplying (3.12) by P_j and integrating gives, for $j \geq 0$,

$$\frac{1}{2^d} \sum_{|k|<2^{d+1}} Q_d(k/2^d) \int_{-1}^{1} u(x - k/2^d) P_j(x)\, \mathrm{d}x = \delta_{dj} \frac{2}{2d+1}. \tag{9.4}$$

Thus the sampled deconvolution row is biorthogonal, through the truncated translate moments, to every Legendre mode. Equation (5.5) is the corresponding block-to-block form after a second synthesis is inserted.

10 Even Christoffel–Darboux projectors as dual translate frames

10.1 The even projection kernel

Define

$$K_N^{\mathrm{e}}(x, y) = \sum_{n=0}^{N} \frac{4n+1}{2} P_{2n}(x) P_{2n}(y). \tag{10.1}$$

It is the kernel of the orthogonal projector onto

$$E_N = \operatorname{span}\{P_0, P_2, \ldots, P_{2N}\}. \tag{10.2}$$

If K_d denotes the full Christoffel–Darboux kernel through degree d, then

$$K_N^{\mathrm{e}}(x, y) = \frac{1}{2}\big(K_{2N}(x, y) + K_{2N}(x, -y)\big). \tag{10.3}$$

Deconvolve in the first variable:

$$K_N^{\sharp}(t, y) = \sum_{n=0}^{N} \frac{4n+1}{2} Q_{2n}(t) P_{2n}(y). \tag{10.4}$$

Theorem 10.1 (Finite translate factorization of the even projector; specialized here). *Put $m = 4^N$. Then*

$$\boxed{K_N^{\mathrm{e}}(x, y) = \frac{1}{m} \sum_{|k|<2m} K_N^{\sharp}(k/m, y) u(x - k/m).} \tag{10.5}$$

Consequently, if

$$g_k(x) = u(x - k/m), \qquad \psi_k(y) = \frac{1}{m} K_N^{\sharp}(k/m, y), \tag{10.6}$$

then

$$\boxed{(\Pi_N^{\mathrm{e}} f)(x) = \sum_{|k|<2m} g_k(x) \langle f, \psi_k \rangle .} \tag{10.7}$$

Proof. For fixed y, the function $x \mapsto K_N^{\mathrm{e}}(x, y)$ has degree $2N$. Its deconvolution is exactly $K_N^{\sharp}(\cdot, y)$, so theorem 2.1 gives (10.5). Integrate against $f(y)$ to obtain (10.7). □

The atoms g_k do not lie in E_N individually, so this is not an ordinary basis expansion of the polynomial subspace. It is a finite synthesis-analysis factorization of its orthogonal projector: the analysis functions ψ_k are even polynomials, while the synthesis functions are compactly supported translates.

Corollary 10.2 (Trace and Hilbert–Schmidt identities; new here)**.** *The families in* (10.6) *satisfy*

$$\sum_{|k|<2m} \langle \psi_k, g_k \rangle = N+1, \tag{10.8}$$

$$\sum_{|k|,|\ell|<2m} \langle g_k, g_\ell \rangle \langle \psi_k, \psi_\ell \rangle = N+1. \tag{10.9}$$

All inner products are over $[-1,1]$.

Proof. Write (10.7) as $\Pi_N^{\mathrm{e}} = \sum_k g_k \otimes \psi_k$. The trace of a rank-one operator $g \otimes \psi$ is $\langle \psi, g \rangle$, while $\operatorname{tr} \Pi_N^{\mathrm{e}} = \operatorname{rank} \Pi_N^{\mathrm{e}} = N+1$, proving (10.8). Expanding the Hilbert–Schmidt norm gives the double sum in (10.9); an orthogonal projector has squared Hilbert–Schmidt norm equal to its rank. □

These identities give exact global diagnostics for any numerical realization of the projector. Unlike pointwise collocation checks, they test the complete interaction between the truncated translate Gram matrix and the deconvolved Legendre analysis Gram matrix.

11 Factorial deconvolution and a sharp central-coefficient law

11.1 Imported asymptotic input

Let

$$C = \mathsf{M}(\pi \mathrm{i}) = \prod_{j=1}^{\infty} \frac{\sin(\pi/2^j)}{\pi/2^j}. \tag{11.1}$$

The nearest poles of $1/\mathsf{M}$ are $\pm 2\pi\mathrm{i}$. The recent repository loop report derives

$$Q_{2n}(0) = \frac{2(-1)^n}{C} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2})\right). \tag{11.2}$$

In particular,

$$|Q_{2n}(0)|^{1/(2n)} \sim \frac{2n}{\pi \mathrm{e}}, \qquad \frac{|Q_{2n}(0)|^{1/n}}{n^2} \to \frac{4}{\pi^2 \mathrm{e}^2}. \tag{11.3}$$

The same report proves that

$$c_n^{(0)} := \frac{u_n Q_{2n}(0)}{4^n} \tag{11.4}$$

is unbounded in absolute value, ruling out atomwise flattening.

11.2 Legendre coefficients cannot decay geometrically

Lemma 11.1 (Sharp root obstruction from nonanalyticity; new here)**.** *The Legendre coefficients of* u *satisfy*

$$\boxed{\limsup_{n\to\infty} |u_n|^{1/n} = 1.} \tag{11.5}$$

Proof. Absolute convergence implies $u_n \to 0$, hence the limsup is at most one. Suppose it were $L < 1$. Choose a Bernstein ellipse parameter $\rho > 1$ with $L\rho^2 < 1$, and then choose q so that $L < q < \rho^{-2}$. Eventually $|u_n| \le q^n$. On every compact subset of the interior of that ellipse, the classical complex bounds for Legendre polynomials give $|P_{2n}(z)| \le C_K \rho^{2n}$. Hence $\sum u_n P_{2n}(z)$ converges locally uniformly and defines a holomorphic extension of u to a neighborhood of $[-1,1]$. This contradicts the nowhere-analyticity of the Fabius–Rvachev function [1]. Thus the limsup is one. □

Theorem 11.2 (Sharp root-limsup of the repeated central atom; new here)**.** *For the coefficient* (11.4),

$$\boxed{\limsup_{n\to\infty} \frac{|c_n^{(0)}|^{1/n}}{n^2} = \frac{1}{\pi^2 \mathrm{e}^2}.} \tag{11.6}$$

If

$$V_n := \frac{|u_n|}{4^n} \sum_{|k|<2\cdot 4^n} |Q_{2n}(k/4^n)| \tag{11.7}$$

is the raw coefficient variation of the n*th block, then*

$$\limsup_{n\to\infty} \frac{V_n^{1/n}}{n^2} \geq \frac{1}{\pi^2 \mathrm{e}^2}. \tag{11.8}$$

Proof. From (11.4),

$$\frac{|c_n^{(0)}|^{1/n}}{n^2} = |u_n|^{1/n} \frac{|Q_{2n}(0)|^{1/n}}{4n^2}.$$

The second factor tends to $(\pi^2 \mathrm{e}^2)^{-1}$ by (11.3); the first has limsup one by lemma 11.1. This proves (11.6). Since the $k = 0$ term occurs in the absolute sum defining V_n, $V_n \geq |c_n^{(0)}|$. □

The theorem is substantially stronger than mere unboundedness. Along a subsequence, the nth root of the repeated central coefficient grows like $n^2/(\pi^2 e^2)$. Thus the canonical coefficient rows are superexponential along a subsequence, while their synthesized functions have L^2 norm

$$\|B_n\|_2 = |u_n| \sqrt{\frac{2}{4n+1}} \longrightarrow 0. \tag{11.9}$$

The closed loop is therefore an extreme cancellation mechanism.

n	atoms	$\|u_n\|$	V_n	$V_n/\|u_n\|$
0	3	5.000×10^{-1}	1.500	3.00
2	63	3.667×10^{-1}	9.057	24.7
4	1023	8.573×10^{-2}	45.549	531
6	16383	2.679×10^{-2}	230.706	8.61×10^3
8	262143	1.243×10^{-2}	7921.35	6.37×10^5
9	1048575	2.446×10^{-3}	37572.4	1.54×10^7

Table 4: Full-grid double-precision scans of the exact rational coefficient polynomials. The raw variation grows while the synthesized mode shrinks.

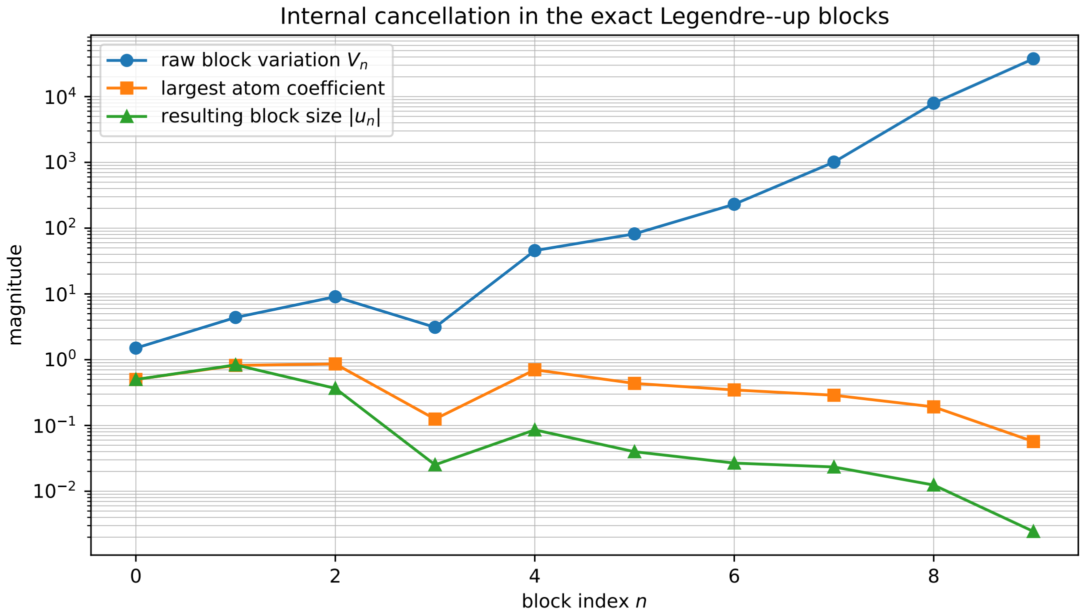

Figure 4: Internal coefficient variation, largest individual coefficient, and resulting block size. The rigorous asymptotic statement is (11.6); the plot only illustrates the pre-asymptotic regime.

11.3 Lambert-W threshold

Solving the leading relation $|Q_{2n}(0)| \asymp (2n/(\pi e))^{2n}$ for the first index at which it exceeds X gives the imported threshold

$$n_Q(X) = \frac{\log X}{2W(\log X/(\pi e))} + \mathcal{O}(1). \tag{11.10}$$

Here the principal branch W appears because one inverts factorial growth. This is distinct from the lower real branch W_{-1} governing inversion of the small-argument endpoint asymptotics of F. The two occurrences share the same algebraic core, inversion of a quantity of the form $y \log y$, but encode different geometries.

12 Connections across the Fabius–Rvachev corpus

12.1 Bell and Bernoulli polynomials

The coefficient pipeline is

$$B_{2r} \longrightarrow \kappa_{2r} \longrightarrow \beta_{2r} \longrightarrow A_n(x) \longrightarrow Q_d(x) \longrightarrow Q_d(k/2^d).$$

The first arrow is the dyadic cumulant factor $(1-4^{-r})^{-1}$; the next two are complete Bell-polynomial transforms; the last two are finite basis changes and dyadic sampling. Bell and Bernoulli polynomials therefore do not merely repackage the result: they are the exact algebra that deconvolves the up probability law.

12.2 q-binomial coefficients and dyadic arithmetic

The coefficient u_n is a finite transform of dyadic Fabius values by (4.4). Those values have q-binomial and q-Pochhammer formulas in the repository. Substitution into (6.9) creates a positive outer series whose inner data are finite q-hypergeometric sums. This separates two roles:

- q-binomial arithmetic computes the spectral weights u_n exactly;

- the dyadic zero multiplicities of $\widehat{u}$ determine which sampled translate lattices reproduce the corresponding polynomial modes.

A useful future objective is to simplify their composition before squaring in (6.9).

12.3 Thue–Morse and nullspace geometry

Thue–Morse signs appear in exact dyadic evaluation of F and in the annihilators used to compress active up-atoms. The new interlevel relation (8.3) is a different but compatible nullspace element: it arises by subtracting two exact Poisson reproductions. For dyadic refinement $r = 2^s$, its periodic mask $1 - r\mathbf{1}_{r|k}$ should be compared with the cyclotomic/q-integer factors

$$\prod_{j=1}^{d}[2^j]_z, \qquad [N]_z = 1 + z + \cdots + z^{N-1}, \tag{12.1}$$

that generate the repository's local polynomiality defects. Their precise factorization relationship is an open algebraic problem stated below.

12.4 Infinite sinc products and energy

The same product (2.3) has three simultaneous meanings:

1. it is the characteristic function of the random series (2.5);
2. its integer zero multiplicities create exact polynomial reproduction;
3. its squared modulus integrates to the positive energy (6.5).

Thus the finite shifted-polynomial theorem and the global L^2 norm are not separate phenomena. Both are consequences of the same sinc product, one through its zeros and the other through its magnitude.

13 Numerical and exact verification

13.1 Exact arithmetic checks

The accompanying script uses Python's arbitrary-precision integers and `Fraction`. Moments are generated from Bernoulli cumulants; reciprocal MGF coefficients use an exact exponential-generating-function recurrence; Legendre and deconvolved polynomials are manipulated coefficientwise; dyadic values of u are evaluated from the finite Thue–Morse/moment formula.

Degree d	$m = 2^d$	atoms	dyadic test points	max exact residual
0	1	3	9	0
1	2	7	17	0
2	4	15	17	0
3	8	31	17	0
4	16	63	17	0
5	32	127	17	0
6	64	255	17	0

Table 5: Exact verification of (3.12).

Two further checks return the rational number zero:

- the complete P_4 quarter-grid null train $m = 16 \to 64$ at seventeen dyadic points;
- the $N = 1 \to 2$ lifting law, first coefficientwise and then as a function at seventeen dyadic points.

No floating approximation enters those tests.

13.2 Reproduction command

From the artifact directory, run

```
python legendre_rvachev_experiments.py
```

The program regenerates all CSV files and figures. Useful options are

```
python legendre_rvachev_experiments.py --no-plots
python legendre_rvachev_experiments.py --max-spectral-mode 120
```

The full-grid coefficient scan is floating point because the $n = 9$ block already contains $1{,}048{,}575$ atoms; its polynomial and spectral coefficients are nevertheless generated exactly before conversion.

14 Conjectures and frontier directions

Conjecture 14.1 (Legendre coefficient saddle law)**.** *Away from a possible sparse set of near-cancellation indices, the coefficients u_n obey a dyadic saddle asymptotic of the form*

$$\log |u_n| = -\frac{(\log n)^2}{2\log 2} + \mathcal{O}(\log n \log\log n), \tag{14.1}$$

with a bounded or slowly varying log-periodic correction inherited from the binary endpoint phase of F.

The same quadratic-log scale occurs in the Fourier-decay and endpoint asymptotic parts of the corpus. A proof should combine a Hilb or Debye approximation for P_{2n} with the lower-branch Lambert-W expansion of the flat endpoint layer. It would turn the root-limsup theorem (11.6) into a full asymptotic for the canonical central atom.

Conjecture 14.2 (Exterior side-lobe explosion)**.** *Extend the finite train B_n from* (4.5) *to all of $\mathbb{R}$ by the same formula and call it $\widetilde{B}_n$. Although $\widetilde{B}_n = u_n P_{2n}$ on $[-1, 1]$, conjecturally*

$$\frac{\left\|\widetilde{B}_n\right\|_{L^2(\mathbb{R}\setminus[-1,1])}}{\|B_n\|_{L^2[-1,1]}} \longrightarrow \infty. \tag{14.2}$$

A stronger version predicts superexponential growth along a subsequence.

This would make the local nature of exact polynomial reproduction visible in physical space: enormous exterior side lobes would cancel only on the target interval.

Problem 14.3 (Optimal null-space gauge)**.** *At a fixed fine scale, add arbitrary vectors from the complete dyadic alias nullspace to the canonical sampled coefficient row. Determine the unique minimum-ℓ^2 row, the minimum-ℓ^1 variation, and their asymptotics for P_{2n} and S_N. Can a multilevel choice of gauges make the lifting updates numerically stable without destroying exactness?*

The root-limsup theorem applies to the canonical sampled gauge. Whether every exact literal-shift representation must exhibit comparable growth is a deeper question about the smallest singular values of the local synthesis map.

Problem 14.4 (Cyclotomic factorization of refinement nulls)**.** *Express the coefficient sequence in* (8.3)*, for $r = 2^s$, in the repository's q-integer/Thue–Morse nullspace basis. Determine whether the refinement cocycle* (8.7) *corresponds to a factorization or filtration of $\prod_{j=1}^{d}[2^j]_z$ by cyclotomic shells.*

Problem 14.5 (Tight or well-conditioned projector frames)**.** *The dual families g_k, ψ_k in* (10.6) *factor an orthogonal projector but are highly redundant and poorly conditioned. Find a nullspace gauge, a nonuniform subset of centers, or a weighted inner product in which the same exact projector has near-tight frame bounds. The trace and Hilbert–Schmidt identities* (10.8)–(10.9) *provide invariant constraints for this optimization.*

Conjecture 14.6 (Irrationality of the Fabius square energy)**.** *The constant* $A_2 = \int_0^1 F(x)^2\,\mathrm{d}x$ *is irrational.*

Even this first arithmetic classification appears nontrivial. The positive rational series, sinc-product integral, and dyadic-Fabius square series supply three very different starting points; transcendence is a natural stronger question only after irrationality is understood.

14.1 Higher-dimensional and other orthogonal systems

The logical pattern is not specific to Legendre polynomials. Whenever a compactly supported atom reproduces a polynomial space and a target function has an orthogonal polynomial expansion, every spectral mode can be replaced by a finite atom train. What is special here is the simultaneous exactness of all arithmetic ingredients: rational moments, dyadic zero multiplicities, Bell–Bernoulli deconvolution, and compact support. Promising extensions are:

- Jacobi and Gegenbauer systems, with endpoint weights chosen to match differentiated energy forms;
- tensor-product and box-spline analogues of the recent multivariate up-like constructions [12];
- spherical harmonics combined with radial compactly supported atoms;
- nonstationary subdivision atoms whose zero multiplicities grow by a prescribed arithmetic schedule.

15 Formalization roadmap

The new results divide naturally into finite algebra and standard Hilbert-space arguments.

1. **Polynomial layer.** At checkpoint `1f9ce6fd9`, `RvachevMomentAppell.lean` formalizes the full rational raw moments, their formal reciprocal/Bell sequence, the rational and real monic reciprocal-moment Appell families, and coefficientwise polynomial deconvolution with exact smoothing recovery and a degree-nonincrease bound. The analytic reciprocal-MGF/differential-series display, specialized parity, explicit low coefficients, and a specialized reciprocal recurrence remain outside that API.
2. **Synthesis and Legendre specialization.** At checkpoint `c51a41fcf`, `RvachevPolynomialSynthesis.lean` formalizes the global and finite sufficient-mesh forms of (2.14), and `FabiusLegendreTranslateBlocks.lean` packages (3.12) and (3.17) at meshes 2^d and 4^n. Minimality and sharpness of those meshes and the closed derivative/Gegenbauer/Appell forms of Q_d remain targets.
3. **Orthogonal block layer.** Complete orthogonality of the polynomial-form blocks, the coefficient Parseval identity, and the exact tail are formalized in `FabiusLegendreEnergy.lean`. The same checkpoint's `FabiusLegendreTranslateBlocks.lean` now formalizes the literal finite up-translate realization, its complete block orthogonality, and the atom-Gram formula. A named outer translate-block `HasSum` and the fixed-scale partial-sum synthesis remain outside these APIs.
4. **Energy layer.** The real-variable equality (6.3) is formalized in `FabiusLegendreEnergy.lean`. At checkpoint `9d5f41c2c`, `FabiusLegendreRationalEnergy.lean` also formalizes its positive rational monotone partial sums, convergence of their real casts to A_2, and all four exact values in (6.6). At checkpoint `b9b240bc0`, `FabiusSquareEnergyFourier.lean` formalizes the full and positive-half-line Fourier energies and closes (6.4)–(6.5) with its four-theorem API.
5. **Refinement layer.** Define finite index embeddings E_r and prove (8.3) by subtraction. The lifting theorem then reduces to coefficient normalization and linearity.
6. **Endpoint layer.** Differentiate the finite sum, use flat support at ± 1, and import the closed endpoint derivatives of P_d.

7. **Projector layer.** Formalize finite-rank operators $g \otimes \psi$, traces, and Hilbert–Schmidt norms to obtain corollary 10.2.
8. **Asymptotic layer.** Reuse the existing theorem for $Q_{2n}(0)$; prove that geometric Legendre decay yields analytic extension; conclude the root-limsup law (11.6).

The interlevel identities and orthogonal block theorem should be relatively short formal developments because their analytic content is already isolated in existing repository theorems. The only new complex-analytic lemma needed for the sharp central law is the standard implication from geometric Legendre coefficient decay to holomorphic extension in a Bernstein ellipse.

16 Conclusion

The requested finite representation is completely explicit. For degree d, one applies the reciprocal moment operator $\mathscr{D}_u = \mathsf{M}(D)^{-1}$ to P_d, samples the resulting polynomial Q_d at the grid $2^{-d}\mathbb{Z}$, and attaches those samples to finitely many literal translates of u. For the even modes in the canonical expansion, the grids are $4^{-n}\mathbb{Z}$ and nest by quarter-grid refinement.

Substitution closes the loop exactly, but its correct unit is a finite cancellation block. The new orthogonality theorem shows that these blocks are not arbitrary: they are a complete orthogonal resolution of u in $L^2[-1,1]$. This yields exact tails, atom-Gram identities, rank-one detail projectors, and an infinite hierarchy of positive spectral energies. The simplest energy is the square integral of the Fabius function, simultaneously a positive rational Legendre series and an infinite sinc-product integral.

The nested grids introduce a second structure. Refining an exact polynomial coefficient row adds a computable null train. Refining a Legendre partial sum adds that invisible gauge correction plus precisely one new orthogonal mode. This is an exact lifting law rather than a numerical analogy.

Finally, the loop is intrinsically ill-conditioned in its canonical atom coordinates. Factorial moment deconvolution combines with the absence of any geometric Legendre decay to force the repeated central coefficient to have root-limsup $n^2/(\pi^2 e^2)$. The function-level series is exceptionally well behaved; the atom-level flattening is exceptionally badly behaved. Any stable computational realization must therefore preserve the spectral block structure, exploit nullspace gauges, or redesign the dictionary rather than merely reorder the individual translates.

A Exact recurrences used by the program

The moments satisfy the cumulant recurrence

$$\mu_0 = 1, \qquad \mu_n = \sum_{j=1}^{n} \binom{n-1}{j-1} \kappa_j \mu_{n-j}. \tag{A.1}$$

The reciprocal-MGF coefficients satisfy

$$\beta_0 = 1, \qquad \beta_n = -\sum_{j=1}^{n} \binom{n}{j} \mu_j \beta_{n-j}. \tag{A.2}$$

The Legendre recurrence is

$$(d+1)P_{d+1}(x) = (2d+1)xP_d(x) - dP_{d-1}(x). \tag{A.3}$$

After each P_d is constructed, the finite operator (2.12) produces Q_d, and (4.2) produces u_n.

For exact dyadic verification, the program uses, for integers $n \geq 0$ and $0 \leq a \leq 2^n$, the finite formula

$$F(a/2^n) = \frac{2^{-n(n+1)/2}}{n!} \sum_{j=0}^{a-1} (-1)^{s_2(j)} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r} (2a-2j-1)^{n-2r}, \tag{A.4}$$

where $s_2(j)$ is the binary digit sum. Then

$$u(q/2^n) = F(1 - |q|/2^n) \tag{A.5}$$

inside the support and is zero at or beyond its endpoints.

B Artifact inventory

The accompanying ZIP archive contains:

- `legendre_rvachev_self_reconstruction.tex` and its compiled PDF;
- `legendre_rvachev_experiments.py`, fully commented and executable offline;
- `README.txt` with reproduction instructions;
- `CORPUS_AUDIT.md`, `REPOSITORY_SNAPSHOT.txt`, and `SHA256SUMS.txt` documenting provenance and package integrity;
- exact CSV checks for Legendre synthesis, the quarter-grid null train, and the lifting step;
- exact Legendre/deconvolution coefficients and positive rational energy partial sums;
- the full-grid coefficient-conditioning table;
- all PDF and PNG figures used in the report.

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